Jar Assignment

1)First Converted the xlsx files to CSV files

```python
'''
Decide to use a programatic way to convert excel files into CSV files instead of third party websites because I am assuming
the data is confidential and propreitory
'''

import pandas as pd
from pathlib import Path



# this is the function to convert the excel files into csv files
def convert_to_csv(path : str, file_name:str):
    dataframe = pd.read_excel(path)

    dataframe.to_csv(f"{file_name}.csv", index=False, header=True)


if __name__ == "__main__":

    #path of the excel files
    project_dir = Path(__file__).parent.parent

    list_of_orders_path = project_dir / "List_of_Orders.xlsx"
    order_details_path = project_dir / "Order_Details.xlsx"
    sales_target_path = project_dir / "Sales_target.xlsx"


    convert_to_csv(list_of_orders_path, "list_of_orders")
    convert_to_csv(order_details_path, "order_details")
    convert_to_csv(sales_target_path, "sales_target")
```

TASK 1 : Sales Analysis

a) Sales and Profitability Analysis

```python
#merge the list_of_orders and order_details on the key 'Order ID'

import pandas as pd
from pathlib import Path



#merges the two csv files and creates a new file
def merge():
    dir = Path(__file__).parent.parent
    list_of_orders_path = dir / "list_of_orders.csv"
    order_details_path = dir / "order_details.csv"

    list_of_orders_df = pd.read_csv(list_of_orders_path)
    order_details_df = pd.read_csv(order_details_path)

    merged_df   = pd.merge(list_of_orders_df, order_details_df, on='Order ID')
    merged_df.to_csv(Path(__file__).parent / "merged_list_of_orders_and_order_details.csv",index=False)


if __name__ == "__main__":
    merge()
```

Here I have merged the required CSV files into one file to work on it as per the requirements

```python
import pandas as pd
from pathlib import Path

#find the total sales amount grouped by category
def total_sales_per_category(dataframe):
    total_sales_per_category = dataframe.groupby('Category')['Amount'].sum()
    return total_sales_per_category

#for each category, find the average profit per order, profit margin as percentage of Amount
def average_profit_and_margin(dataframe):
    result = dataframe.groupby('Category').agg(
        average_profit = ("Profit","mean"),
        total_amount = ("Amount","sum"),
        total_profit = ("Profit",sum)
    )
    print(type(result))
    result["profit_margin"] = (result["total_profit"] / result["total_amount"]) * 100

    return result


if __name__ == "__main__":
    dir = Path(__file__).parent.parent
    dataframe = pd.read_csv(dir / "merged_list_of_orders_and_order_details.csv")


    total_sales_per_category = total_sales_per_category(dataframe)
    print(total_sales_per_category)


    average_profit_and_margin = average_profit_and_margin(dataframe)
    print(average_profit_and_margin)
```

Have used groupby method to group by the category and then used aggregate methods to find the sum and mean

```
op/JarAssignment/Task1/SubTask1/main.py
Category
Clothing       139054
Electronics    165267
Furniture      127181
Name: Amount, dtype: int64
<class 'pandas.DataFrame'>
             average_profit  total_amount  total_profit  profit_margin
Category
Clothing          11.762908        139054         11163       8.027817
Electronics       34.071429        165267         10494       6.349725
Furniture          9.456790        127181          2298       1.806874
```

As we can see from the output, clothing and electronics are the high performing categories and furniture is a low performing category. We can clearly see that clothing and electronics have a higher profit margin around 8% and 6.3% respectively whereas Furniture has only about 1.8%.
We can see througout the data that the total profit and the average profit in Clothing and Electronics are higher compared to furniture

There could be several reasons why this is the case, we could assume that the cost of making the electronics and the clothing items are cheaper, they are getting sold more and hence their total profit and the amount sold is higher. We can also assume that clothes and electronics have so much sales because its easier to give discounts and offers while still making sure that the profit margins are higher. Clothes and electronics are also frequently bought compounding this effect of making them profitable

On the other hand furniture cost of making is higher, higher cost of logistics and also marketing it to a customer is more difficult as most of the indian users do not buy furniture as often. And also giving discounts and offers may hurt the profit margins

TASK 1 : Sales Analysis

b) Target Achievement Analysis

```python
import pandas as pd
from pathlib import Path


def target_change():
    dir = Path(__file__).parent.parent
    df = pd.read_csv(dir / "sales_target.csv")

    # get the data from the furniture category
    furniture = df[df["Category"] == "Furniture"].copy()

    # converting the column to datetime
    furniture["Month of Order Date"] = pd.to_datetime(
        furniture["Month of Order Date"]
    )

    # sorting it from january up to december
    furniture = furniture.sort_values("Month of Order Date")

    furniture["Percentage Change"] = (
        furniture["Target"].pct_change() * 100
    )

    return furniture


if __name__ == "__main__":
    result = target_change()
    print(result)
```

Reads the csv, and copy's only the data relevant to the furniture category and then sort using the dates given and use the pct_change method to see the month-on-month change

```
Month of Order Date   Category  Target  Percentage Change
         2026-01-19  Furniture   11500                NaN
         2026-02-19  Furniture   11600           0.869565
         2026-03-19  Furniture   11800           1.724138
         2026-04-18  Furniture   10400         -11.864407
         2026-05-18  Furniture   10500           0.961538
         2026-06-18  Furniture   10600           0.952381
         2026-07-18  Furniture   10800           1.886792
         2026-08-18  Furniture   10900           0.925926
         2026-09-18  Furniture   11000           0.917431
         2026-10-18  Furniture   11100           0.909091
         2026-11-18  Furniture   11300           1.801802
         2026-12-18  Furniture   11400           0.884956
venv) PS C:\Users\tauro\OneDrive\Desktop\JarAssignment>
```

Overall the target has been stable but we can see that during the month of april the target was decreased significantly, around 11.86%.

There are certain startegies so that the targets and the actual performance of the category(Furniture in this case) are close.

- We can use historical data to see the month-on -month sales of furnitures and analyse on which months the sales are greater or lesser and to identify seasonal patterns and set realistic targets.

- If there is a huge mismatch between the target and the performance of the product then we can reduce the target and slowly increase it.

As we can see in the data both of these strategies were applied, there must have been a mismatch and the target was reduced and later it was increased slowly looking at the historical perfomances of the product

TASK 1 : Sales Analysis

c) Regional Performance Insights

```python
import pandas as pd
from pathlib import Path

def top_states():
    dir = Path(__file__).parent.parent
    df = pd.read_csv(dir / "merged_list_of_orders_and_order_details.csv")

    # counting the orders of each state
    order_count = (
        df.groupby("State")["Order ID"]
        .nunique()
        .sort_values(ascending=False)
    )

    #get the top 5 states
    top_5_states = order_count.head(5).index

    # create dataframe of states present in top 5 states
    top_5_df = df[df["State"].isin(top_5_states)]

    result = (
        top_5_df.groupby("State")
        .agg(
            order_count=("Order ID", "nunique"),
            total_sales=("Amount", "sum"),
            average_profit=("Profit", "mean")
        )
        .sort_values("order_count", ascending=False)
    )

    return result

if __name__ == "__main__":
    results  = top_states()
    print(results)
```

again using the groupby method and aggregate methods to calculate the required metrics

State	order_count	total_sales	average_profit
Madhya Pradesh	101	105140	16.326471
Maharashtra	90	95348	21.296552
Rajasthan	32	21149	16.986486
Gujarat	27	21058	5.344828
Punjab	25	16786	-10.150000

From the output we can observe that Pubjab and Gujarat despite having 25 and 27 sales respectively have a poor average profit compared to Rajastan that has around 32 sales comparable to the other two states doing an average profit of 16.98

So Pubjab and Gujarat are the states that are needed to be prioritised for further improvement

Github link : https://github.com/JeethanTauro/Jar-Assignment

Question 2 : (10 marks)

App Exploration:

Explore the features and user experience of the Jar app. Highlight five things you found particularly effective and user-friendly. Additionally, identify five areas where improvements could be made, providing your reasoning for each suggestion.

5 Features I found effective and user friendly:

1) Catering to the indian audience by making helpful videos in hindi
2) Instant withdrawal and savings
3) The future planning feature where we can set goals
4) Dailing savings using Autopay feature
5) The "Gold price drop" feature at the top (its helpful in knowing that the user can buy gold at a cheaper price)

All these features are very useful especially the autopay feature, i dont have to constantly set a reminder for myselft to save, and also daily, weekly and monthly savings options are good too.
Another feature I feel is a top feature is the instant withdrawal and instant savings The option to withdraw my money and invest instantly is really user friendly

5 areas where I felt there is room for improvement:

1) The cluttered home page: The home page is clutered with "spin and win", "weekly magic", "everyday rewards", "coupons", "success stories" etc
I personally feel that the home page should have only the neccessary things such as withdrawal, instant save (which is there) and tracking progress, showing the current price of gold. I feel these other things that are there makes the home page more cluttered and can distract the user or just make it a bit annoying to use. I clicked on my profile and that page was more useful as i could access everything what i want from there itself be it changing my savings, withdrawing, everything is well organised. My suggestion would be shifting the quick actions reward stuff from the home page to rewards sections anf the success stories can be in a seperate testomonials page

2) The unskipable tutorial: I think the videos are really great and well animated but there is a problem, even though i have watched it once, it keeps showing the same tutorial, when i click on "change savings" it gives me an almost 6-7 sec preview or a tutorial on what the "change savings" does, but as a user i think showing it once as default is good but then it becomes annoying as users can't skip it.

3) Gamification of the app: Well I do like rewards, and coupons as a user but i feel it was a bit too much for a first time user, i have invested 50 rs today and then I got rewards stuffs, coupons, points , spin and win etc etc. I couldnt figure out which reward would give me which benefit? if i spin what do i actually win? what can i do with the coupons?. So i would suggest keeping the rewards a bit secondary. Maybe there could be some user guide to guide through the rewards section because there are bunch of different things, like buy gold with points, buy items with points, wiin some coupon and then applying the coupon code and buying gold etc So a good guide maybe in hindi with some animations could help a first time user like me to navigate through these. (This was a personal take because I have never used things like coupons and rewards and stuff except applying coupons automatically while ordering food)

4) Detailed description of the savings: I have invested a total of 50 rs, but while withdrawing I could see that there's actually only 43 rs , basically the gold price has gone down, so my investment has gone down with gold (but thats not the issue) the issue is , on the homepage it still shows that i have invested 50rs which is true but i feel it should also show the current value of my investment, as the price of gold changes and my investment value increases or decreases it must show that too

5) Beginner Friendly Guide: As a first-time investor, I understood that I was saving money, but I wasn't immediately clear on how my savings were converted into gold, how the value of my investment changes with the gold price, and what happens when I withdraw. Since this involves real money, a short beginner-friendly walkthrough showing the complete journey from saving to gold investment to withdrawal would make the experience much more reassuring.

Question 3: (10 marks)

Product Exploration:

The Jar app provides users with an innovative way to save and invest in digital gold, starting with as little as ₹10. It automates savings and investments, making financial planning seamless and accessible. As the first Made-in-India app to pioneer such a solution, Jar has successfully created a niche in automated savings and investment. Building on its strong foundation and leveraging its existing user base and trust, what are some new business opportunities Jar could venture into, to expand its offerings and enhance user engagement? Discuss how the app can utilize its strengths, such as automation, a user-friendly design, and established credibility, to seamlessly integrate these new services, deepen its value proposition, and achieve significant milestones in the financial ecosystem.

1. Problem I noticed

Currently, Jar mainly focuses on helping users save and invest, especially in digital gold. However, when users actually want to spend their money, they have to move to other UPI/payment apps like PhonePe, Google Pay or Paytm. This creates a separation between spending and saving. The user spends money on one platform and then has to consciously come back to Jar to save or invest. This means Jar has an opportunity to become a more regular part of the user's everyday financial activity instead of being used only when the user wants to save or invest.

2. Opportunity / Gap

Jar already has some of the most important things required user trust, a large existing user base, a simple user-friendly interface. The opportunity is to connect the user's daily spending with their saving and investment behaviour. Instead of competing only as another payment app, Jar can differentiate itself by making every payment contribute towards the user's financial goals. This would allow Jar to move from being just a savings/investment app to becoming a more complete everyday financial platform.

3. Feature I am proposing is "Jar Pay"

I propose introducing a Jar Pay feature where users can maintain a balance in a Jar wallet and use it for their everyday payments. For example, if a user adds ₹1,000 to the wallet, they can use that amount for their daily transactions. The interesting part is that Jar can connect these payments with its existing savings model. For example, if the user spends ₹500 and has selected a 5% automated savings rule, ₹25 can automatically be invested into digital gold. The user can also receive cashbacks and rewards for making payments. So instead of the user having to separately think about spending, rewards and investing, the entire process happens together of spending, saving and investing in gold

5. Benefits to Jar

For Jar, this can significantly increase user engagement and retention. Currently, a user may open Jar when they want to save or invest, but payments can make Jar part of their everyday routine. If users make multiple transactions through Jar every day, the number of interactions with the platform increases significantly. This also gives Jar more opportunities to encourage users to invest and use other financial products in the future.

6. Overall Value Proposition

The bigger idea is that Jar can move from simply being an app where users save money to an app that helps users spend, earn and save at the same time. The payment feature becomes the entry point, while automated investing remains the core of Jar's identity. This allows Jar to build a stronger relationship with its existing users and gradually expand into a larger financial ecosystem while still staying true to what made the platform successful in the first place.